Southern New Hampshire University

CS-360: Assignment 1-4: Assessment of "Phyphox" mobile phone application

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1 Introduction

Technology is the application of scientific principles to solve problems.[1] Technology can be relatively simple, like a campfire to keep warm when outside. This paper concerns itself with *information technology*, which is all about inputting data, processing it with formal procedures, and finally outputting its results and computations. These information systems have taken the form of an abacus, nomograms,[2] "looming machines," and most recently, electronic devices that stores information in terms of digital impulses.

In the latter case of electonic computers, one of its strength is that they can be compacted down into smaller pieces of hardware that can store and compute more data than their predecessors.[3] A recent kind of electronic

[1]This definition is paraphrased from Dictionary.com: https://www.dictionary.com /browse/technology (accessed on Sept. 6, 2026).

[2]Something of a "graphical computer," consisting of an alignment chart mapping an input value to its respective precomputed outputs.

[3]Moore's law posits that the number of transistors in microchips doubles every two years (2), while at the same time, the cost of computers his halved; see https://www.intel.com/content/www/us/en/newsroom/resources/moores-law.html (ac-

computer is the smartphone, or the mobile phone. These "mobile phones" are a fusion of traditional cell phones with more general computing platform that are similar to personal computers. These smartphones are slighly bigger than a typical cellphone, but can still fit into the pockets of a typical end-user's trousers.

Mobile phones solve specific software engineering problems through "apps," which are packages of software that allow the software engineers to extend the functionality of mobile phones. Like how the purist mathematicians work on maths problems for the sake of proving correctness without regard for real world applications, apps can be written "for the sake of technology." But more often than not, apps are written to solve a problem posed by the end-user. These problems can regard long-distance communication with other end-users, advertising, navigation, or any kind of problem involving computation or communication.[4] Here, I intend to explore one such app that assists students of basic physics with experimental research.

2 Case Study: Phyphox

Phyphox (n.d.) is an app targeting both the Apple iOS and Android platforms that allows for end-users to perform cheap, cost-effective experiments in basic physics. To study this app, I have installed the Android version of it onto an *Android Studio* emulator, and while doing research for this project, I identified an undergraduate textbook (Organtini, 2021) which documents the Phyphox app, along with other tools for scientific experiment. Figure 1 shows Phyphox running under Android studio.

The Phyphox app lists number of physics experiment related tools under the categories "raw sensors," "acoustics," "camera," "everyday life," "mechanics," "timers," and general "tools." Most of these tools use the mobile phone's sensors to measure physical phenomena and then analyse said data, though there are some tools that are used for simulating physical phenomena.[5]

From a cursory inspection of Phyphox's interface, I can describe the "'look and feel" as simple: with black-greyish and orange backgrounds and white, foreground colours. Each tool is put in its respective category. For each

cessed on Sept. 6, 2026).

[4]It should be noted that information systems cannot solve all problems, and may sometimes end up creating more problems than what is solved. In some cases, a simpler technology or solution is superior to a complicated information system.

[5]Such as in the case of the "tone generator," which generates a tone of a specific frequency.

Figure 1: Phyphox running under *Android Studio*

tool, a button is rendered with the name of the tool and a description of it is displayed. On the far-left of the button, a small icon is displayed, presumably so that the end-user can quickly identify the tool when needed.

Regarding specific problems that Phyphox can solve, it is primarily used for data collection and analysis. Some of the measuring tools built into the app include:

- **Audio Autocorrelation**: to "measure the frequency of a single tone."
- **Location (GPS)**: to "get raw position data from satellite navigation."
- **Roll**: determines the velocity of the mobile phone that is running the app, after "plac[ing it] in a roll."
- **Audio Spectrum**: to "display the frequency spectrum of an audio signal."
- **Acoustic Stopwatch**: to "get the time between two acoustic events."

Regarding the "raw sensors," these tools may not necessarily be applied for scientific or engineering purposes. "Location (GPS)" could be used for travel and navigation and the various "stopwatch" tools can be used for measuring distance traveled. But for the most part, these tools are for the systematic study of natural phenomena, and may be "overkill" for solving everyday problems that can be solved with more commercial apps like *Google Maps* or those "health apps" for measuring step count and kilometers walked.

3 Discussion

Phyphox mainly solves problems in measurement and analysis of physical science experiments, and it target audience appears to be undergraduate students of the physical sciences, engineers and applied scientists who are running experiments to test their inventions, and lay-persons interested in autodidactic science the. In the latter case of "lay-persons," they may casually download the app to solve a specific problem that they are faced with, and may not regularly use it as frequently as the students, engineers, or committed autodidact.

Unlike most mobile phone apps published onto the Apple App Store or Google Play, Phyphox is a non-profit effort. The institution who published the app are a university, and their goals are not to make a profit, but to democratize education and to make science education more affordable and accessable. The creators of Phyphox do not employ much propaganda to

convince end-users to use download and use their app, but instead, the app is "advertised" through word of mouth, and by being included in scientific and engineering curricula and technical textbooks.[6]

3.1 Acknowledgements

Before writing up this report, I used the Claude large language model when doing research for a personal research project outside of this course. Specifically, I was looking for science experiments to generate datasets that can be used as input to estimate the parameters of a theoretical probability distribution.

The Claude model printed out *Phyphox* as one, of many, research tools that can be used in "kitchen table" experiments. I worked with this Claude model just about a week before working on this assignment, and I do not have any previous experience with Phyphox,[7] so I chose this app to study from the "outsider" perspective, hopefully with minimal vernacular biases but enough subject matter know-what to approach the subject of study. Though interestingly, I have the undergraduate physics experiment textbook written by Organtini (2021) in my library, but have not had the chance to read it and do its scientific experiments, so the Phyphox app is still "new to me."

References

CS-340 (n.d.). *CS-360 Module One Assignment Guidelines and Rubric.* Southern New Hampshire University.

Organtini, G. (2021). Physics Experiments with Arduino and Smartphones. In Undergraduate Texts in Physics. Springer International Publishing. https://doi.org/10.1007/978-3-030-65140-4

Phyphox (n.d.). *Physical Phone Experiments.* Retrieved from: https://phyphox.org/

[6]Such as the undergraduate physics experiment textbook by Organtini (2021).

[7]At least not that I can remember. I may have used Phyphox a long time ago in the past, but it was "out of sight" long enough to be "out of mind."